



HR ANALYTICS

# HR Analytics Project

## Employee Attrition Analysis & Risk Identification

Turning workforce data into clear insights to help identify attrition risk and support better retention decisions.

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HR ANALYTICS

# Problem Statement

Understanding why employees leave is essential to reducing turnover and protecting performance.

1

## What's happening?

Companies face frequent employee attrition.

2

## Consequences

High recruitment cost, loss of experience, productivity decline.

3

## Core question

Why do employees leave the company?

PROJECT GOAL

# Project Objective



## Analyze employee behavior

Identify attrition drivers

## Detect high-risk employees early

Spot patterns before turnover happens

## Support HR decision-making with data

Enable more informed, timely action

# Dataset Overview

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## Demographics

Age, Gender, Education



## Job Info

Department, Role, Job Level



## Salary

Monthly Income, Salary Hike



## Satisfaction

Job, Environment, Work-Life  
Balance



## Experience

Years at company, Promotion  
history



## Target

Attrition (Yes / No)

# Technical Workflow

A clean, professional four-phase process flowing downward from preparation to reporting.

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**Data Prep**

**EDA**

**Modeling**

**Dashboards**



# Python Data Cleaning

Essential preprocessing steps using Python libraries:



**Remove unused columns**



**Drop ID fields**



**Standardize text values**



**Remove duplicates**



**Export cleaned dataset**



```
3]: # Drop useless columns  
df = df.drop(columns=["EmployeeCount", "StandardHours", "Over18"])  
  
# Drop id  
df = df.drop(columns=["EmployeeNumber"])  
  
# Clean text  
df["BusinessTravel"] = df["BusinessTravel"].str.replace("_", " ")  
  
# Remove duplicates  
df = df.drop_duplicates()
```



```
4]: # save the file  
df.to_excel("HR_Cleaned_1.xlsx", index=False)
```



# Excel Data Transformation

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Main Tasks:



**Created Age Groups**



**Created Income Categories**



**Converted ratings into labels**



**Prepared data for dashboard visualization**



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 Copy  
 Format Painter  
Clipboard

Aptos Narrow 11 A<sup>^</sup> A<sup>v</sup>  
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|                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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|  Paste<br>✕<br> Cut<br> Copy<br>✕<br> Format Painter | <p>Aptos Narrow 11 A<sup>^</sup> A<sup>v</sup></p> <p><b>B</b> <i>I</i> <u>U</u> ▾ ▢ ▾  ▾  ▾</p> | <p>≡ ≡ ≡  ▾ &gt;¶ ▾ ab Wrap Text</p> <p>≡ ≡ ≡    Merge &amp; Center ▾</p> <p>Clipboard Font Alignment</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

AH2 ▾ : ✕ ✓ *fx*

=IF(S2<5000,"Low",IF(S2<10000,"Mid",IF(S2<15000,"High","Very High")))

# Why Data Cleaning Matters?

## Impact of Data Preparation:

Improved data quality

More accurate analysis

Better visualization

Easier segmentation & insights

**Output: Cleaned dataset ready for analysis**



# SQL Data Analysis

Database queries to extract and analyze HR patterns



Database queries to extract and analyze HR patterns:

- Extract HR insights
- Aggregate workforce data
- Analyze attrition patterns
- Support KPI calculations

## Key SQL Operations

WHERE

GROUP BY

COUNT() / AVG()

CASE WHEN

```
SELECT
  department,
  COUNT(*) AS employee_count,
  AVG(salary) AS avg_salary
FROM employees
WHERE status = 'Active'
GROUP BY department;
```

# Key Findings from SQL

---

## Key Insights:

- Sales department showed the highest attrition
- Employees under 30 had higher turnover
- Overtime employees were more likely to leave
- Lower job satisfaction increased attrition risk

## SQL Supported:

**Attrition Analysis**

**Salary Analysis**

**Workforce Segmentation**

**Employee Risk Detection**



File Edit View Git Project Tools Extensions Window Help Search

master Execute

Object Explorer

Connect

DESKTOP-QLGH9L7 (SQL Server 17.0.1110.1 - DESKTOP-QLGH9L7)

- Databases
- Security
- Server Objects
- Replication
- Always On High Availability
- Management
- SQL Server Agent
- XEvent Profiler

HR\_data.sql - ...l store (55)

```
1 SELECT
2     COUNT(*) AS Total_Employees
3 FROM hr_data;
4 SELECT
5     ROUND(
6         (SUM(CASE WHEN Attrition = 1 THEN 1 ELSE 0 END) * 100.0)
7         / COUNT(*),
8     2) AS Attrition_Rate
9 FROM hr_data;
10 SELECT
11     Department,
12     COUNT(*) AS Attrition_Count
13 FROM hr_data
14 WHERE Attrition = 1
15 GROUP BY Department
16 ORDER BY Attrition_Count DESC;
17 SELECT
18     CASE
19         WHEN Age < 30 THEN 'Under 30'
20         WHEN Age BETWEEN 30 AND 40 THEN '30-40'
21         WHEN Age BETWEEN 41 AND 50 THEN '41-50'
22         ELSE 'Above 50'
23     END AS Age_Group,
24     COUNT(*) AS Attrition_Count
25 FROM hr_data
26
27
28
```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF Windows 1252

Connected. (1/1) DESKTOP-QLGH9L7 (17.0 RTM) DESKTOP-QLGH9L7\origin... master 00:00:00 Row: 0, Col: 0 0 rows

# Data Model & DAX Measures

Understanding the foundational data model and key analytical measures is crucial for accurate attrition analysis.

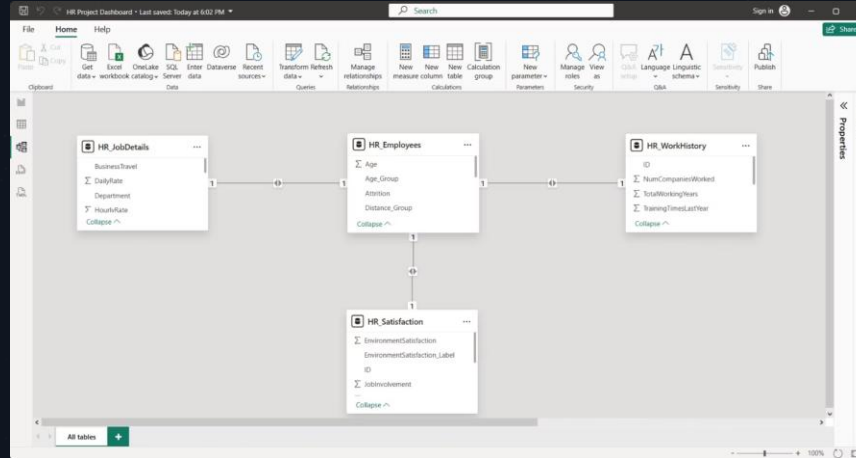


Table Relationships:



## Key DAX Measures:

These DAX measures were developed to calculate attrition rate, assess individual employee risk, categorize risk levels, and project the impact of salary increases on attrition reduction.

### 1. Attrition Rate

```
DIVIDE([Attrition  
Count], [Total  
Employees], 0)
```

### 2. Attrition Risk Score

```
VAR OverTimeScore =  
IF(MAX(HR_JobDetails[Ov  
erTime]) = "Yes", 30,  
0)  
VAR SatisfactionScore =  
IF(MAX(HR_Satisfaction[  
JobSatisfaction]) <= 2,  
25, 0)  
VAR BalanceScore =  
IF(MAX(HR_Satisfaction[  
WorkLifeBalance]) <= 2,  
20, 0)  
VAR DistanceScore =  
IF(MAX(HR_Employees[Dis  
tanceFromHome]) >= 20,  
15, 0)  
VAR StockScore =  
IF(MAX(HR_JobDetails[St  
ockOptionLevel]) = 0,  
10, 0)  
RETURN OverTimeScore +  
SatisfactionScore +  
BalanceScore +  
DistanceScore + StockScore
```

### 3. Risk Category

```
IF([Attrition Risk  
Score] >= 51, "High  
Risk",  
IF([Attrition Risk  
Score] >= 21,  
"Medium Risk", "Low  
Risk"))
```

### 4. Projected Attrition Reduction

```
[Attrition Rate] *  
(1 - 'Salary  
Increase %'[Salary  
Increase % Value] /  
200)
```

Query 1 Mohamed Nasser Corr and Reg... x

```

190  (
191      STDDEV(YearsWithCurrManager) * STDDEV(Attrition)
192  )
193  AS corr_YearsWithCurrManager|
194  FROM new_schema.`hr_cleaned_1(sheet1)`;
195  -- Regression part
196  • SELECT
197  -- Age

```

- new\_schema
  - sakila**
    - Tables
    - Views
    - Stored Procedures
    - Functions
  - sys
  - world

◀ ▶ |   | Jump to ▼

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

| Result Grid          |                        | Filter Rows:          | Export:                  | Wrap Cell Content: |
|----------------------|------------------------|-----------------------|--------------------------|--------------------|
| Standard_Age         | StandardBusinessTravel | StandardDailyRate     | StandardDistanceFromHome | StandardEducation  |
| -0.05854571285727213 | 0.03137013226939844    | -0.020833083795354302 | 0.028655453759570564     | -0.0115369744      |



Result Grid

| Result 1 | Result 2 | Result 3 | Result 4 |
|----------|----------|----------|----------|
| 1        | 2        | 3        | 4        |

Read Only Context Help Snippets

### Output

#### Action Output

| # | Time       | Action                                                                                                | Message           | Duration / Fetch      |
|---|------------|-------------------------------------------------------------------------------------------------------|-------------------|-----------------------|
| ✓ | 1 18:09:40 | SELECT ( AVG(Age * Attrition) - AVG(Age) * AVG(Attrition) ) / (STDDEV(Age) * STDDEV(Attritio...       | 1 row(s) returned | 0.047 sec / 0.000 sec |
| ✓ | 2 18:09:40 | SELECT -- Age ( ( AVG(Age * Attrition) - AVG(Age) * AVG(Attrition) ) / ( AVG(Age * Age) - POW...      | 1 row(s) returned | 0.015 sec / 0.000 sec |
| ✓ | 3 18:09:40 | SELECT ( AVG( ( JobSatisfaction - stats.avg_js ) / stats.std_js ) * Attrition ) - AVG( JobSatisfac... | 1 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ | 4 18:09:40 | SELECT / " Age " / ( AVG( (Age - stats.avg_Age) / stats.std_Age ) * Attrition ) - AVG((Age - stat...  | 1 row(s) returned | 0.063 sec / 0.000 sec |

### Activate Windows

Go to Settings to activate Windows.

Query Completed



# Hypothesis (1)

## 1. Job Difficulty and Attrition (Positive Relationship) (+)

We assume a positive relationship between job-related difficulties and employee attrition. Variables such as **travel frequency** and **distance between home and work** increase the likelihood of attrition.

## 2. Adaptation and Stability Factors (Mixed Relationships)

### A. Employment History (Positive Effect on Attrition) (+)

Employees who have worked in a higher number of companies (greater job mobility/turnovers) are more likely to experience attrition

### B. Stability and Organizational Attachment (Negative Effect on Attrition) (-)

The following factors are expected to reduce the likelihood of attrition:

- Age (older employees tend to be more stable in the labor market)
- Years at the company (higher organizational attachment)
- Performance rating (strong performance increases retention incentives)
- Job involvement



## Hypothesis (2)

- Job level (higher positions reduce willingness to take risk and change jobs)
- Job satisfaction
- Relationship satisfaction
- Work–life balance
- Training received in the last year (reflecting both skill development and organizational investment)

### **3. Financial Incentives (Negative Relationship) (-)**

Higher financial incentives are expected to reduce the likelihood of employee attrition (The variables are Monthly Income, Stock Option Level, Percent Salary Hike).

# Results (1)

- We conducted **correlation, regression, and standardization analyses** on the dataset.
- As hypothesized at the beginning, all hypotheses were verified and found to be supported by the results.
- We used regression analysis in addition to correlation analysis because we aim to examine **causal relationships and the magnitude of their effects** (i.e., coefficients in the regression table).
- Since **monthly income is measured in thousands**, unlike the other variables, its inclusion in regression without adjustment could distort the results. Therefore, we applied **standardization to all variables** to ensure a balanced and comparable dataset.
- It is noted that the coefficients are relatively small. This may be explained by several factors:
  1. Employee attrition is influenced by many variables. We use 15 variables and there are also beyond those included in this model.
  2. The analysis captures general relationships, which may become stronger when further segmented by variables such as gender, educational background, age groups, and departments.

## Results (2)

- According to the results, **travel frequency** appears to be the strongest factor increasing the likelihood of attrition.
- On the other hand, **monthly income** is the most influential financial factor, while **age** and **job level** are the most important factors, related to stability and adaptation, associated with reducing attrition.

| Factors related to attrition |                               | Regression   | Correlation | Standardization |
|------------------------------|-------------------------------|--------------|-------------|-----------------|
| 1                            | Distance From Home            | 0.004        | 0.078       | 0.029           |
| 2                            | Travel Frequency              | 0.040        | 0.085       | 0.031           |
| 3                            | No of Companies worked before | 0.006        | 0.043       | 0.016           |
| 4                            | Age                           | -0.006       | -0.159      | -0.059          |
| 5                            | Years at Company              | -0.008       | -0.134      | -0.049          |
| 6                            | Performance Rating            | 0.003        | 0.003       | 0.001           |
| 7                            | Jobs Involvement              | -0.067       | -0.130      | -0.048          |
| 8                            | Job Level                     | -0.056       | -0.169      | -0.062          |
| 9                            | Job Satisfaction              | -0.035       | -0.103      | -0.038          |
| 10                           | Relationship Satisfaction     | -0.016       | -0.046      | -0.017          |
| 11                           | Work Life Balance             | -0.033       | -0.064      | -0.024          |
| 12                           | Training Times Last Year      | -0.017       | -0.059      | -0.022          |
| 13                           | Monthly Income                | -1.24893E-05 | -0.160      | -0.059          |
| 14                           | Stock Option Level            | -0.059       | -0.137      | -0.050          |
| 15                           | Percent Salary Hike           | -0.001       | -0.013      | -0.005          |

# Power BI Dashboard Overview

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## Key Performance Indicators:

1,470

Total Employees

1,233 active

\$6,503

Average Monthly Income

37

Average Age

productive workforce stage

2.69

Avg Previous Companies

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## Dashboard Features:



Real-time KPI monitoring and trend tracking



Employee segmentation by department, role, and demographics



Interactive filtering for drill-down analysis



Attrition tracking and risk identification

# Dashboard Insights — Overview Page

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## Critical Findings:

**16.12%**

### Attrition Rate

1 out of every 6 employees resigns — a high turnover rate requiring immediate attention.

**2.73**

### Low Employee Satisfaction

Company-wide issue across all departments, not isolated to a single area.

R&D department: 961 employees (largest) HR department: 63 employees (smallest). This makes R&D the most critical department in any business decision.

**215**

### Promotion Gap

No promotion for 5+ years, with highest concentration among Sales Executives (63 employees). Potential source of future attrition.



1.47K

Total Employees

37

Avg Age

6.50K

Avg Monthly Income

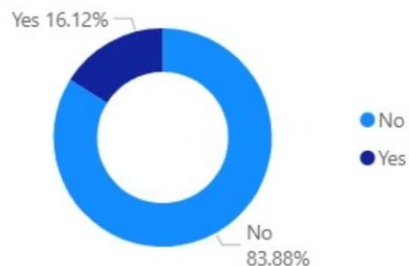
1.23K

Active Employees

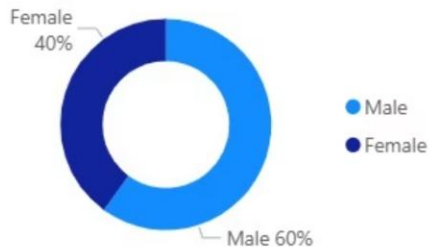
3

Avg Num Companies Worked

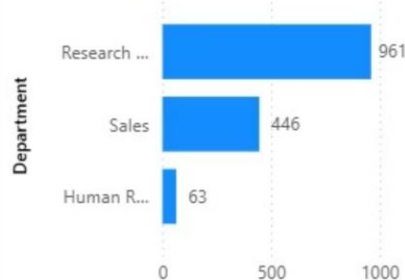
Total Employees by Attrition



Total Employees by Gender



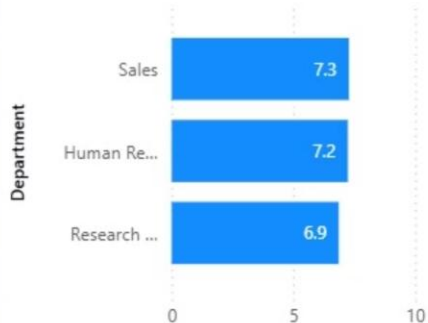
Total Employees by Department



Age\_Group

- ☐ Select all
- ☐ +46
- ☐ 18-25
- ☐ 26-35
- ☐ 36-45

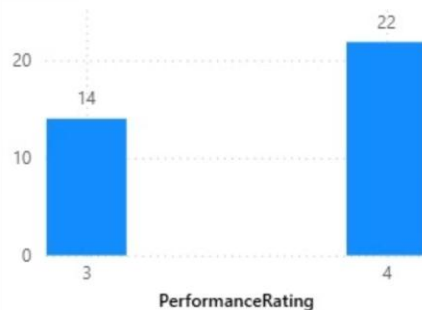
Avg Years At Company by Department



Avg Job Satisfaction by Department



Avg Percent Salary Hike by PerformanceRating



| ID   | Department             | JobRole       |
|------|------------------------|---------------|
| 46   | Research & Development | Research Dir  |
| 124  | Research & Development | Research Dir  |
| 127  | Research & Development | Healthcare R  |
| 179  | Sales                  | Sales Executi |
| 238  | Sales                  | Manager       |
| 426  | Research & Development | Manager       |
| 702  | Sales                  | Manager       |
| 753  | Research & Development | Laboratory T  |
| 862  | Sales                  | Manager       |
| 915  | Research & Development | Healthcare R  |
| 927  | Sales                  | Sales Executi |
| 938  | Research & Development | Manager       |
| 977  | Research & Development | Manufacturi   |
| 585  | Research & Development | Manager       |
| 654  | Research & Development | Manager       |
| 923  | Research & Development | Manager       |
| 1079 | Research & Development | Research Dir  |
| 1157 | Research & Development | Manufacturi   |
| 1222 | Research & Development | Healthcare R  |

# Dashboard Insights — Attrition & Salary Analysis

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## Key Findings:

### Significant Salary Gap by Role

**Managers:** \$17,182 vs **Sales**

**Representatives:** \$2,626

555% salary difference explains  
39.8% attrition rate among Sales  
Representatives.

### Age-Salary Relationship

Salary increases with age until age  
40, then declines regardless of  
company tenure.

Critical retention issue for  
experienced employees over 40.

### Stock Options Impact on Attrition

**No Stock Options:** 24.4% attrition  
risk (highest)

**Medium Stock Options:** 7.6%  
attrition risk (lowest)

Financial benefits are key retention  
drivers.

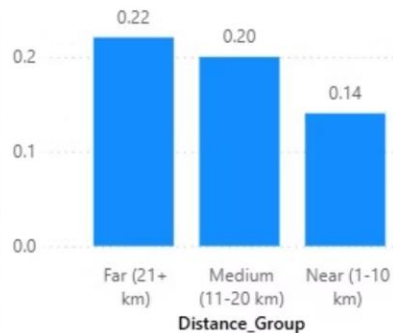
16.12%

Attrition Rate

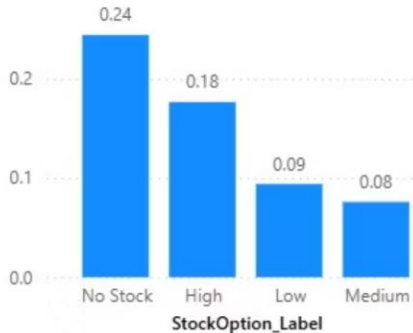
2.73

Avg Job Satisfaction

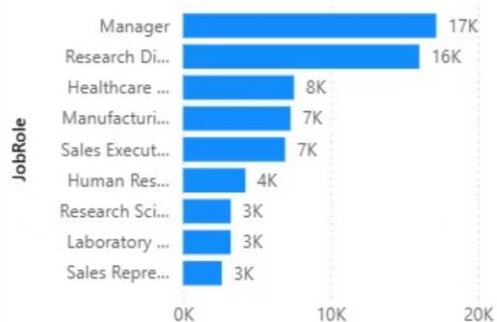
Attrition Rate by Distance\_Group



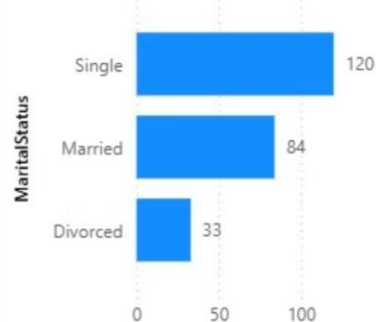
Attrition Rate by StockOption\_Label



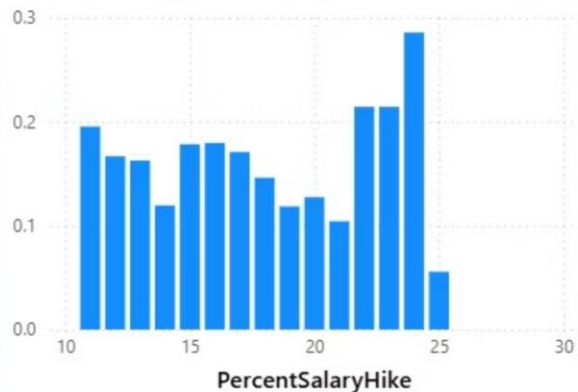
Avg Monthly Income by JobRole



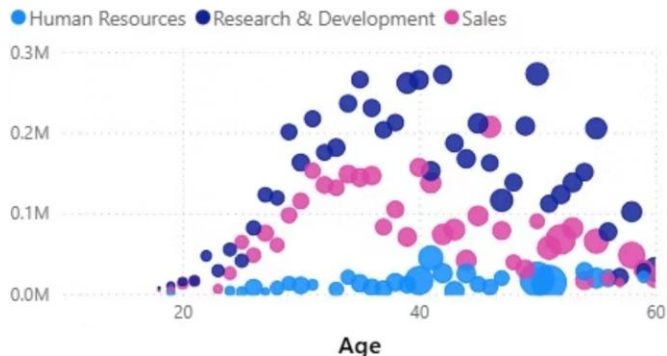
Attrition Count by MaritalStatus



Attrition Rate by PercentSalaryHike



Sum of MonthlyIncome and Average of YearsAtCompany by Department and Age



Average of JobSatisfaction by Attrition and Salary\_Groups



# Predictive Analysis & Forecasting

## Salary Growth Trajectory

Salary increases with overall work experience, rising from \$1,524 to \$6,623 during the first 9 years. Forecast predicts continued growth, reaching up to \$14,730 with extended tenure.

## Employee Satisfaction Plateau

Employee satisfaction remains almost constant between 2.59 and 2.92 regardless of years of experience. Indicates company is not improving work environment over time — a structural issue requiring intervention.

## Salary Increase Impact Simulation

If salaries increase by 20%:

- Projected income: \$6,503 → \$7,804
- Attrition rate: 16.12% → 15%
- At-risk employees: 237 → 213

Enables management to make data-driven decisions for employee retention.

## High-Risk Employee Identification

High-risk employees have simultaneously:

- Overtime work
- Low job satisfaction
- No stock options

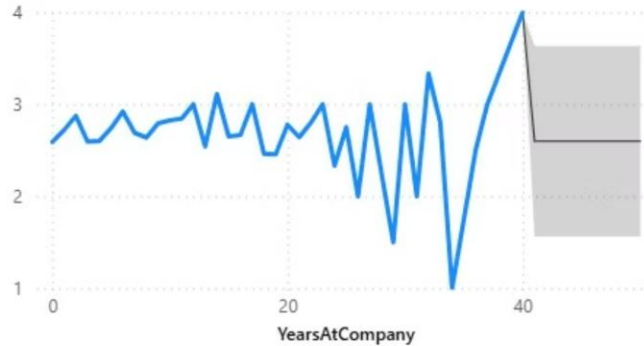
The dashboard identifies these employees by name, enabling HR to intervene before resignation.

Salary Increase %

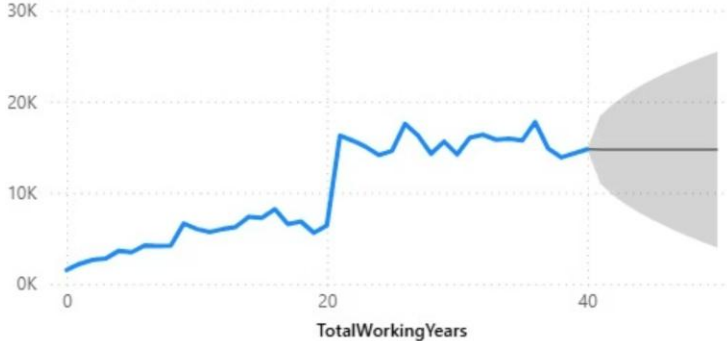


Extra Years

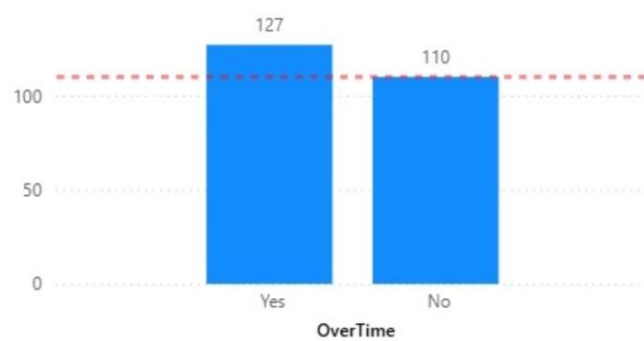
Avg Job Satisfaction by YearsAtCompany



Avg Monthly Income by TotalWorkingYears



Attrition Count by OverTime



| otion | Attrition Risk Score | Risk Category |
|-------|----------------------|---------------|
| 0     | 100                  | High Risk     |
| 0     | 100                  | High Risk     |
| 2     | 100                  | High Risk     |
| 1     | 90                   | High Risk     |
| 9     | 90                   | High Risk     |
| 5     | 90                   | High Risk     |
| 7     | 90                   | High Risk     |
| 0     | 85                   | High Risk     |
| 1     | 85                   | High Risk     |
| 0     | 85                   | High Risk     |
| 0     | 85                   | High Risk     |
| 2     | 85                   | High Risk     |
| 13    | 85                   | High Risk     |
| 0     | 85                   | High Risk     |
| 7     | 85                   | High Risk     |
| 5     | 85                   | High Risk     |
| 0     | 85                   | High Risk     |
| 0     | 85                   | High Risk     |
| 0     | 85                   | High Risk     |
| 2     | 85                   | High Risk     |
| 0     | 85                   | High Risk     |
| 0     | 80                   | High Risk     |
| 2     | 80                   | High Risk     |
| 2     | 80                   | High Risk     |
| 0     | 80                   | High Risk     |
| 1     | 80                   | High Risk     |
| 1     | 80                   | High Risk     |
| 10    | 80                   | High Risk     |
| 1     | 80                   | High Risk     |
| 0     | 75                   | High Risk     |
| 0     | 75                   | High Risk     |
| 1     | 75                   | High Risk     |
| 1     | 75                   | High Risk     |
| 7     | 75                   | High Risk     |
| 45    | Medium Risk          | Medium Risk   |

7.80K

Projected Income

14.51%

Projected Attrition Reduction

213

Employees At Risk After Raise

8.13K

Projected Income After Experience

# Tableau Dashboard

## Interactive Dashboard Development using Tableau

### Key Features:



#### Dynamic Charts & Filters

Interactive visualizations for real-time data exploration



#### KPI Monitoring

Total Employees (1,470), Total Attrition (237), Attrition Rate (16.12%), Active Employees (1,233), Avg Age (37)



#### Department Analysis

Department-wise attrition breakdown and trends



#### Demographic Insights

Employee segmentation by age group, gender, and job satisfaction



#### User-Friendly Design

Clear, visually appealing format optimized for end-user engagement

### Dashboard Capabilities:

- Drill-down functionality for detailed analysis
- Real-time data updates and trend monitoring
- Comparative analysis across departments and demographics
- Data storytelling through intuitive visualizations
- Interactive filters for custom reporting and insights





# HR Analytics Dashboard

|               |            |             |
|---------------|------------|-------------|
| KPI           | Att.Gender | D.Attrition |
| E.Attrition   | Jop S.     | N.E.Age     |
| Att.Age group |            |             |

|                                                     |
|-----------------------------------------------------|
| Education Field                                     |
| <input checked="" type="checkbox"/> Human Resources |
| <input checked="" type="checkbox"/> Life Sciences   |
| <input checked="" type="checkbox"/> Marketing       |
| <input checked="" type="checkbox"/> Medical         |
| <input checked="" type="checkbox"/> Other           |

Σ Total Employees

1,470

➡ Total Attrition

237

⚙️ Attrition Rate

16.12%

⏻ Activate Employees

1,233

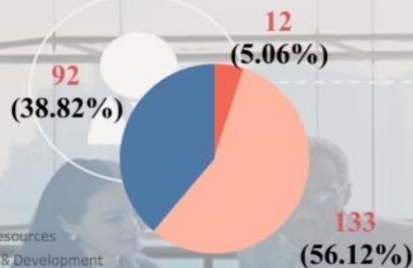
👤 Avg. Age

37

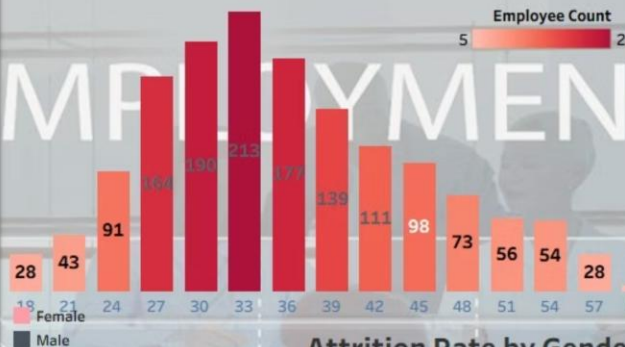
Attrition by Gender



## Department Wise Attrition



## No of Employee by Age Group



## Jop Satisfaction Rating

| Job Role                  | Job Satisfaction |     |     |     | Total |
|---------------------------|------------------|-----|-----|-----|-------|
|                           | 1                | 2   | 3   | 4   |       |
| Human Resources           | 10               | 16  | 13  | 13  | 52    |
| Research Director         | 15               | 16  | 27  | 22  | 80    |
| Sales Representative      | 12               | 21  | 27  | 23  | 83    |
| Manager                   | 21               | 21  | 27  | 33  | 102   |
| Healthcare Representative | 26               | 19  | 43  | 43  | 131   |
| Manufacturing Director    | 26               | 32  | 49  | 38  | 145   |
| Laboratory Technician     | 56               | 48  | 75  | 80  | 259   |
| Research Scientist        | 54               | 53  | 90  | 95  | 292   |
| Sales Executive           | 69               | 54  | 91  | 112 | 326   |
| Total                     | 289              | 280 | 442 | 459 | 1,470 |

## Education Field Wise Attrition



## Attrition Rate by Gender for different Age Group

